

VIVA

IP Addressing & Networking Basics

Q1. What is an IP address?

Ek unique number jo har device ko network me identify karta hai.

Q2. What is the difference between Private and Public IP addresses?

Private IP local networks me use hota hai (e.g., 192.168.x.x), Public IP internet pe globally unique hota hai.

Q3. What is Subnetting?

Network ko chhote logical segments me todna taaki performance aur security improve ho.

Q4. How do you calculate the number of hosts in a subnet?

Formula: $2^n - 2$, jahan n host bits hain.

Q5. What is a Router?

Device jo networks ke beech data packets forward karta hai aur best path choose karta hai.

Q6. What is DHCP?

Protocol jo automatically IP addresses assign karta hai clients ko.

Q7. What is DNS and how does it work?

Domain ko IP me convert karta hai using DNS resolver, root, TLD, aur authoritative server.

Server & OS Administration

Q8. What is one main difference between Linux and Windows Server?

Linux open-source hai, Windows Server proprietary hai.

Q9. Give one advantage and one disadvantage of Linux.

Advantage: Stable & secure. Disadvantage: Steeper learning curve.

Q10. Give one advantage of Windows Server.

Easy GUI management aur Microsoft ecosystem integration.

Q11. Why would you choose Linux for a corporate network?

Cost effective, secure, aur customization-friendly.

Network Security & Monitoring

Q12. What is IPsec?

Protocol suite jo IP packets ko encrypt aur authenticate karta hai.

Q13. Name two main protocols used in IPsec.

AH (Authentication Header) aur ESP (Encapsulating Security Payload).

Q14. Why is Network Monitoring important?

Issues detect karna, performance maintain karna, security threats identify karna.

Q15. Name two network monitoring tools.

Nagios, Wireshark.

Linux System Administration

Q16. Command to add a new user in Linux?

```
useradd username
```

Q17. Command to create a new group in Linux?

```
groupadd groupname
```

Q18. Command to check running processes in Linux?

```
ps -aux ya top
```

Q19. Command to configure firewall in Linux?

```
iptables ya firewallld
```

Server Configuration

Q20. Default port number of FTP?

21

Q21. What is Samba used for?

Linux aur Windows ke beech file & printer sharing.

Q22. Default server port of DHCP?

UDP 67

Q23. Default port of DNS?

53

Q24. Main config file of Apache web server in Linux?

`/etc/httpd/conf/httpd.conf`

Networking Lab Concepts

Q25. One benefit of VLAN?

Network segmentation aur improved security.

Q26. One disadvantage of Static Routing?

Manual updates required, large networks me difficult.

Q27. Maximum hop count of RIP?

15

Q28. Purpose of ACL (Access Control List)?

Network traffic ko allow/deny karne ke rules define karna.

Q29. Difference between manageable and non-manageable switch?

Manageable configurable hota hai, monitoring support karta hai; non-manageable plug-and-play hota hai.